

Delta Drills — Mastery Models Evidence Review

Purpose. Lay out the four mastery-estimation models under consideration for the prerequisite-gated bootcamp app, present the underlying mathematics, summarise the two AI-generated reports already in hand, cross-check their claims against the audited primary sources in `papers/mastery-estimation/`, and surface the open decision points. **No recommendation is made here** — this is an evidence dossier for personal evaluation.

Companion document: `Delta-Drills-Current-Model-EWMA.md` (full specification of the engine as deployed today).

0. The four candidate models

In each subsection: per-skill state, update rule, time decay (if any), the form of uncertainty (if any). Notation: s is a skill; n is the attempt index for that skill; $y_n \in \{0, 1\}$ is the correctness of attempt n ; Δt is days elapsed since the previous attempt.

0.1 Exponentially Weighted Moving Average (current Delta Drills stack)

Per-skill state. Two numbers plus a counter: a baseline b_s (EWMA of `grade × difficulty / 100`), a correctness rate p_s (EWMA of `1[grade > 85]`), an attempt count n_s , and a last-update timestamp.

Update. On attempt n with feedback-derived weight $\alpha(n) \in \{0.30, 0.60, 0.85\}$ and difficulty d_n :

$$b_n = \alpha(n) \cdot \frac{\text{grade}_n \cdot d_n}{100} + (1 - \alpha(n)) b_{n-1}, \quad p_n = \alpha(n) \cdot \mathbf{1}[\text{grade}_n > 85] + (1 - \alpha(n)) p_{n-1}$$

Target next difficulty: $b_n \cdot \mu(p_n)$ where μ is the piecewise multiplier capped at 2.5.

Time decay. Yes — multiplicative shrinkage of both b and p toward priors $B_0 = 0$, $P_0 = 0.5$ at half-life $\tau_{1/2} = 14$ days:

$$b \leftarrow B_0 + (b - B_0) \cdot 2^{-\Delta t / \tau_{1/2}}, \quad p \leftarrow P_0 + (p - P_0) \cdot 2^{-\Delta t / \tau_{1/2}}.$$

Uncertainty. None explicit. The values are always defined; there's no signal for "I don't have enough data on this skill."

Multi-tag credit. When a composite drill is rated, each tagged subtopic gets an independent EWMA update (equal-credit averaging).

0.2 Elo rating system

Per-skill state. One number per skill (R_s^{skill}) and one number per student (R^{user}).

Predicted success on an attempt:

$$\hat{y} = \frac{1}{1 + 10^{(R_s^{\text{skill}} - R^{\text{user}})/400}}.$$

Update on observed $y \in \{0, 1\}$:

$$R^{\text{user}} \leftarrow R^{\text{user}} + K \cdot (y - \hat{y}), \quad R_s^{\text{skill}} \leftarrow R_s^{\text{skill}} - K \cdot (y - \hat{y}).$$

The constant K caps how much one attempt can move either rating. Educational Elo typically uses different K values for student and skill, and often a decaying K (large for cold start, smaller as evidence accumulates).

Time decay. Not in standard Elo. Glicko/Glicko-2 extensions add a "rating deviation" (RD) that grows over time, mimicking forgetting; vanilla Elo has none.

Uncertainty. Standard Elo: none. Glicko-2: RD plays this role.

Multi-tag credit. Standard approach: average the per-skill ratings of all tags on the item; update each tag's rating independently from the observed outcome. **This is a known weakness** — a student strong on tag A and weak on tag B gets an averaged prediction that's wrong in both directions. The published fix is **M-ERS (Multidimensional Elo Rating System)** from Park et al. 2019, which maintains a vector of ability estimates rather than a scalar.

0.3 Streak / mastery counter

Per-skill state. One integer: the count of consecutive correct answers, plus optionally a "mastered" flag.

Update.

$$\text{streak}_s \leftarrow \begin{cases} \text{streak}_s + 1 & y_n = 1 \\ 0 & y_n = 0 \end{cases}$$

Declare "mastered" when $\text{streak}_s \geq N$ for some threshold N (Khan Academy historically used $N = 3$ or 5).

Time decay. None native. Some implementations add: "reset to zero if no attempt for T days."

Uncertainty. None.

Multi-tag credit. Each tag carries its own independent streak.

0.4 Bayesian Beta-Bernoulli with time decay

Per-skill state. A Beta posterior parametrized by two counters (α_s, β_s) .

Update on observed $y \in \{0, 1\}$:

$$\alpha_s \leftarrow \alpha_s + y, \quad \beta_s \leftarrow \beta_s + (1 - y).$$

Time decay between attempts (engineering choice, not part of vanilla Beta-Bernoulli): multiplicative shrinkage toward the prior (α_0, β_0) at half-life τ :

$$\alpha_s \leftarrow \alpha_0 + (\alpha_s - \alpha_0) \cdot 2^{-\Delta t/\tau}, \quad \beta_s \leftarrow \beta_0 + (\beta_s - \beta_0) \cdot 2^{-\Delta t/\tau}.$$

Uncertainty. Yes, native: - Posterior mean: $\mu_s = \frac{\alpha_s}{\alpha_s + \beta_s}$ - Posterior precision: $\nu_s = \alpha_s + \beta_s$

(a direct measure of evidence count) - Posterior variance: $\frac{\alpha_s \beta_s}{(\alpha_s + \beta_s)^2 (\alpha_s + \beta_s + 1)}$

Gating typically requires BOTH the mean and the precision to clear thresholds:

$$\text{Ready}(s) \equiv \mu_s \geq \theta_\mu \wedge \nu_s \geq \nu_{\min}.$$

Multi-tag credit. Each tag updated independently (equal credit), same as the current Delta Drills approach.

1. Comparison axes

1.1 Per-skill state size

Model	Numbers stored per skill	Numbers stored per student
Current EWMA	3: b_s, p_s, n_s + timestamp	0
Elo	1: R_s^{skill}	1: R^{user}
Streak	1: streak count	0
Beta-Bernoulli	2: α_s, β_s + timestamp	0

1.2 Cold-start behavior

Model	Behavior with 0 attempts	Behavior with 1 attempt
Current EWMA	Cold-start ladder serves difficulty 25/50/75 for first 3 questions	After 1 attempt, b and p exist but variance is implicit
Elo	Initialized to default rating (typically 1500); first attempt produces useful update	Yes, usable
Streak	0/0; mastery flag off	$\text{streak} = 1$, mastery flag off
Beta-Bernoulli	Prior ($\alpha_0 = 1, \beta_0 = 1$) gives mean 0.5, precision 2	Updated immediately; precision = 3, mean tracks the observed outcome

1.3 Built-in uncertainty signal

Model	Has uncertainty?	What plays that role
Current EWMA	No	—
Elo	No (standard); yes in Glicko/Glicko-2	Rating deviation (RD) in Glicko
Streak	No	—
Beta-Bernoulli	Yes, native	Posterior precision $\nu = \alpha + \beta$

1.4 Time decay / forgetting

Model	Native decay?	Mechanism
Current EWMA	Yes (14-day half-life)	Multiplicative shrinkage of b, p toward priors
Elo	No in standard form	Glicko adds RD that grows over time
Streak	No native; some impls add reset-on-stale	—
Beta-Bernoulli	No native	Engineering: multiplicative shrinkage of α, β toward (α_0, β_0)

1.5 Multi-tag credit assignment

Model	Default approach	Known weakness	Fix in literature
Current EWMA	Equal credit: each tag gets its own independent update	Same as PFA-compensatory but Maier 2021 found equal-credit beat compensatory empirically	None needed per Maier
Elo	Average ratings of tags; update each independently	"Strong-on-A, weak-on-B" wrong-in-both-directions	M-ERS (Park 2019) — multidimensional ability vector
Streak	Independent streak per tag	A student short on one tag never unlocks even if strong on others	—
Beta-Bernoulli	Independent (α, β) per tag	Same as EWMA equal-credit	—

1.6 ARENA-difficulty / ceiling problem

Model	Has the problem?	Why / why not
Current EWMA	Yes — <code>target_difficulty</code> clamped to [0,100]	Global scalar with hard ceiling
Elo	No	Ratings are relative; no fixed ceiling
Streak	No, but for a worse reason	No scale at all — over-promotion failure mode
Beta-Bernoulli	No	Per-skill probability, no global scale

1.7 Interpretability to students

Model	What you can show the student	Honest interpretation
Current EWMA	"72% mastery"	Intuitive but conflates difficulty and accuracy
Elo	"Your rating: 1340, skill rating: 1280"	Opaque without context
Streak	"3 in a row → mastered"	Intuitive but loses partial state

Model	What you can show the student	Honest interpretation
Beta-Bernoulli	"78% likely correct (based on 14 attempts)"	Most honest — explicit confidence

1.8 Implementation cost from current state

Model	Required changes	Risk
Current EWMA	None — already deployed	—
Elo	Replace <code>apply_feedback</code> math; add per-user rating storage; redefine ALL gate thresholds for the new relative scale	Moderate. Migration of existing user state is non-trivial.
Streak	Trivial — counter per skill. But would mean abandoning a lot of nuance.	Low engineering risk, high pedagogical risk
Beta-Bernoulli	Replace <code>apply_feedback</code> math with (α, β) update; add decay; gates need both mean + precision conditions	Moderate. State migration is straightforward (transform existing p and n into compatible (α, β)).

2. Failure modes per model

2.1 Current EWMA

1. **Ceiling effect on `target_difficulty`.** Once a student plateaus on flashcards, `target_difficulty` clamps at 100 and the engine has no way to represent "ready for harder-than-flashcard content." Real, affects ARENA gating.
2. **No native confidence signal.** A student with 3 attempts is treated like a student with 30; the only way to know how thin the evidence is is to read n_s separately.
3. **Two-EWMA aggregation.** Engine stores both b_s and p_s . Gating code uses $p_s \cdot 100$; difficulty selection uses $b_s \cdot \mu(p_s)$. These can disagree (high p with low b = lucky on easy questions; passes gate, shouldn't).

2.2 Elo

1. **No forgetting in standard form.** A skill mastered a year ago is treated identically to one mastered yesterday unless you bolt on Glicko-style RD growth.
2. **Multi-tag averaging is a known weakness** (cf. Park 2019).
3. **K-factor calibration is non-trivial.** Pelánek 2016 specifically tunes separate K values for student vs. skill ratings, and often a decay schedule for K as evidence accumulates. One global K misbehaves at both cold-start and steady-state regimes.
4. **Gate semantics shift entirely.** Current gates like " $p \times 100 \geq 70$ " become " $R^{\text{user}} \geq R_s^{\text{skill}} + \delta$ " — a completely different threshold semantics that needs re-tuning skill-by-skill.

2.3 Streak / mastery counter

1. **Over-promotion failure mode.** A student who gets 3 difficulty-25 cold-start questions right has the same streak as a student who got 3 difficulty-90 questions right. **No difficulty awareness.** This is a *worse* failure than the EWMA's ceiling problem — the model gates students as "ready for ARENA" on the basis of having gotten 3 easy questions correct.
2. **No partial credit / no graduated mastery state.**
3. **No regression handling.** Hard reset on a wrong answer destroys all built-up evidence in one bad attempt.

2.4 Bayesian Beta-Bernoulli

1. **No difficulty awareness in the base form.** Like the streak counter, the (α, β) update treats all correct/wrong answers identically regardless of question difficulty. Mitigation: weight the updates $(\alpha \leftarrow \alpha + w \cdot y)$ where w scales with difficulty — but this is engineering choice not in the audited literature.
 2. **Decay rate has to be set by hand.** Same as the EWMA's 14-day choice — no published study tells you the right number for a bootcamp context.
 3. **Specific architecture (Beta + decay + prereq graph) has thin literature.** One audited paper (Bijl 2025). The underlying Bayesian math is textbook; the deployment in this exact configuration is largely engineering.
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3. Summary of the two AI reports

3.1 Report 1 — "Gap Analysis"

Format. Goes through each model and lists what was missing or under-developed in the prior conversation. Critique-focused, no citations.

Material new contributions over our prior conversation:

1. Reframes streak's failure mode: it's *over-promotion via easy questions*, not ceiling effects. This is correct and important.
2. Catches that the switching cost for Elo is higher than I'd characterized: gate threshold semantics change fundamentally, not just the storage layer.
3. Identifies that NO model addresses prerequisite-graph errors (a cross-cutting blind spot).
4. Raises post-mastery regression UX: what does a student SEE when their mastery slides back down? Each model handles this differently and none was compared.
5. Raises student-facing explainability as a UX dimension never compared across models.
6. Calls out that multi-tag Elo averaging is a published weakness with a published fix (M-ERS / Park 2019).
7. Notes that K-factor calibration depth was hand-waved in the prior comparison.
8. Notes Pelánek 2016 (focused study on Elo for adaptive learning) is a different paper from Pelánek 2017 (broader survey of BKT/PFA/Elo), and they should be cited distinctly.

Factual issues in Report 1:

- Claims a Beta-Bernoulli formula typo of the form `alpha → alpha + beta + (1 if wrong else 0)`. **This formula was never written.** The prior conversation showed the standard `alpha → alpha + 1 (on success), beta → beta + 1 (on failure)`. Report 1's typo allegation is unfounded.
- Attributes a "5+ attempts before estimates settle" claim to EWMA. The number is from Maier 2021's empirical study of Performance Factors Analysis (a different model entirely). The current EWMA settles much faster — with $\alpha = 0.85$ on "A lot" feedback, the EWMA's effective memory length is ~ 1.2 attempts.
- Says "EWMA formula never shared." False as of today — the formulas are fully spelled out in `Delta-Drills-Current-Model-EWMA.md`.

3.2 Report 2 — "Comparative Review"

Format. Survey-style with pros/cons per model, inline citation markers (`【1†L121-L124】` etc.), and a recommendation at the end. Cleaner narrative but less critical.

Material claims in Report 2:

1. Names Pelánek 2016 and 2017 as the Elo sources for educational use.
2. Cites a specific empirical figure: *"Elo's task-difficulty estimates had $r \approx 0.70$ correlation with ground truth after just 5 responses"* — attributed to citation `【16†L47-L52】`.
3. Recommends Elo as the migration target.
4. Identifies Beta-Bernoulli as "principled" but "less field-tested" — accurate.
5. Frames streak as a baseline that more sophisticated models outperform (consistent with Maier 2021's framing).

Issues with Report 2:

- The citation markers (`【1†...】` , `【16†...】` , etc.) are opaque — without the source list it's impossible to tell which paper the " $r \approx 0.70$ after 5 responses" claim originates from. **This number is not directly verifiable from the report alone.** Highly recommend reading Pelánek 2016 §results to confirm whether this claim exists in the original paper at the granularity stated.
- Does not catch the gaps Report 1 catches (streak failure mode reframing, gating-semantics-change, prereq-graph-error blind spot, post-mastery regression, explainability, M-ERS for multi-tag Elo).
- Hand-waves multi-tag aggregation similarly to the prior conversation (no mention of M-ERS / Park 2019, which IS in the audited corpus).
- Claims "moderate work" for Elo migration without addressing the gate-semantics shift.

3.3 Where the two reports agree

- Elo is the strongest single recommendation if a switch is to be made.
- Beta-Bernoulli is principled but under-published in this exact architecture (one paper, Bijl 2025).
- Streak alone is too weak.
- EWMA is functional but has overconfidence/ceiling problems.

3.4 Where the two reports differ

Topic	Report 1	Report 2
Streak's failure mode	Over-promotion via easy questions	Coarse, no partial credit
Switching cost for Elo	"Migration would require redefining every gate threshold"	"Moderate work... retune thresholds for new scale"
Multi-tag Elo	Known weakness, points at M-ERS / Park 2019	Default averaging is fine, no concern raised
Prereq-graph errors	Cross-cutting blind spot, all four models	Not raised
Explainability	Real UX gap	Not raised
Strength of recommendation	None — only lists open questions	Elo, with migration steps

4. Cross-check of specific claims against your audited corpus

The audited papers in `papers/mastery-estimation/` are the primary sources. Below: which report's claims can be checked against which paper, and the current state of verification.

Claim	Source claimed	Audited paper	Verification status
"PFA's compensatory model AUC 0.78; even-skill averaging AUC 0.78; conjunctive AUC 0.67"	Maier 2021	<code>2021_maier-baker-stalzer_pfa-challenges.pdf</code> §4.3	Verified — read end-to-end. Numbers are: compensatory 0.7818, conjunctive 0.6725, even-skill 0.7849.
"PFA needs 5+ practices per (student, skill) for reliable estimates"	Maier 2021	Same	Verified §4.1 — AUC went 0.725 → 0.755 from 2 → 5 practices, plateaued at ~12 practices.
"Math Academy FIRE formula: $\text{repNum} \rightarrow \max(0, \text{repNum} + \text{speed} \cdot \text{decay}^{\text{failed}} \cdot \text{rawDelta})$ and memory decay $(0.5)^{\text{days}/\text{interval}}$ "	Skycak 2023	<code>2023_skycak_fire-spaced-repetition.html</code>	Verified — read end-to-end. Formulas reproduced verbatim from the paper.
"Skycak abandoned Bayesian Knowledge Tracing for Math Academy because complexity exploded"	Skycak 2023	Same	Verified verbatim — direct quote: <i>"complexity explodes when you have to make tons of different kinds of decisions... I ended up</i>

Claim	Source claimed	Audited paper	Verification status
			<i>moving towards less of a probabilistic approach and more of a 'physical' approach."</i>
"Elo's task-difficulty estimates had $r \approx 0.70$ correlation with ground truth after just 5 responses" (Report 2)	Citation [16†L47-L52]	Likely 2016_pelane_k_elo-adaptive-educational.pdf	Not yet verified. Recommend reading Pelánek 2016 §results to confirm the exact figure and the conditions under which it was observed.
"Pelánek 2017 Hypothesis 1: logistic models better for fluency/memory, BKT better for understanding/sense-making"	Pelánek 2017	2017_pelane_k_bkt-logistic-and-beyond.pdf p. 33	Verified via prior audit (the v2 mastery doc cross-checked this and noted the v1 doc misframed it).
"Elo is simple, robust, and effective for adaptive education"	Pelánek 2016	2016_pelane_k_elo-adaptive-educational.pdf abstract	Likely verified (the v2 audit doc confirms abstract quote). Re-reading the §results details is the missing step.
"M-ERS (Multidimensional Elo Rating System) extends Rasch-Elo to multidimensional ability vectors, fixing the multi-tag averaging weakness"	Report 1	2019_park-cornillie_multidim-irt-monitoring-ability.pdf	Not yet verified end-to-end. The v2 audit doc confirms the paper exists and introduces M-ERS; the specific claim about multi-tag handling is consistent with the paper's framing but the implementation details warrant a primary-source read.
"Beta-Bernoulli + decay over a prereq graph is published only in Bijl 2025"	Both reports	2025_bijl_probabilistic-decay-trees.pdf	Partially verified. Audit doc confirms Bijl 2025 uses Beta-distribution basis functions, continuous-variable Dynamic Bayesian Network, explicit decay over a graph of skills. Not yet read end-to-end. Recommended if

Claim	Source claimed	Audited paper	Verification status
			Beta-Bernoulli is seriously considered.
"Streak / three-in-a-row is a baseline more sophisticated models outperform"	Report 2	Maier 2021 mentions this framing in §1	Verified — Maier 2021 explicitly says: <i>"most commercial adaptive learning systems still continue to either use BKT... or simpler heuristics such as three-in-a-row correct."</i>
"Khan Academy used streak-based mastery and moved away from it"	Both	Not in audited corpus	Unverified. Common knowledge in the ed-tech world but no primary-source citation in the corpus.
"Standard Elo K-factor needs separate calibration for student vs skill ratings"	Report 1	Pelánek 2016 / Yudelson 2019	Likely verified (the v2 audit doc notes Yudelson 2019: <i>"Elo, I Love You Won't You Tell Me Your K"</i> is the dedicated K-tuning paper). Yudelson 2019 is in the corpus references but not in the downloaded set — would need to retrieve.

5. Open decision points neither report fully resolves

These are the questions that need answers before code changes are made. Each is followed by where the answer would come from.

1. **Gate threshold values for any chosen model.** For Beta-Bernoulli: what θ_μ and $\nu_{\min}$? For Elo: what rating differential between student and skill counts as "ready"? Neither report quantifies these. **Resolvable only with simulation or first-cohort data.**
2. **K-factor schedule for Elo.** Single value, two values (student vs skill), or decaying with experience? **Pelánek 2016 + Yudelson 2019 cover this directly. Worth reading both before committing.**
3. **Decay half-life for Beta-Bernoulli.** Current EWMA uses 14 days. Same value for Beta? Different? **No published guidance — engineering choice, tunable from data.**
4. **Multi-tag credit for any chosen model.** Equal-credit averaging (current) per Maier 2021 is empirically reasonable. M-ERS upgrade for Elo is published (Park 2019) but adds significant

complexity. **Resolution:** read Park 2019 to assess whether the M-ERS upgrade is worth the complexity.

5. **Prerequisite-graph error resilience.** None of the four models addresses this. The graph itself is an instructor-authored artifact subject to error. **Resolution:** graph-level (instructor review, sample-edge validation), not model-level. **Currently deferred.**
6. **Post-mastery regression UX.** What does the student see when mastery slides back down? Each model produces different behavior; no comparison done. **Resolution:** product-design decision, not model-selection decision.
7. **Migration cost from current EWMA state.** For Elo: how do you initialize $(R^{\text{user}}, R_s^{\text{skill}})$ from existing (b_s, p_s, n_s) ? For Beta: can you transform p_s, n_s into (α_s, β_s) such that the migration preserves user mastery estimates? **Resolution:** needs a 1-page migration spec per chosen target model.
8. **Explainability target.** Do you want students to see a mastery percentage? Does the chosen model support that natively? **Affects choice among Elo (no native percentage) vs. Beta (native posterior probability) vs. EWMA (current 0-100).**
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6. Evidence tier per claim

A rough rubric for how much weight to put on each claim.

Tier	Definition	Examples
Textbook	Foundational math, not in dispute	Beta-Bernoulli conjugacy; Elo logistic prediction; EWMA convergence properties
Peer-reviewed empirical	Published in a venue with peer review, reproduced or replicated	Maier 2021 PFA AUC numbers; Pelánek 2016/2017 surveys of BKT/Elo
Peer-reviewed novel proposal	Published but as a single-paper architecture, not replicated	Park 2019 M-ERS; Bijl 2025 PDT
Production-validated, proprietary	Deployed in real systems but specific implementation not published	Math Academy FRe (formulas public, implementation proprietary)
Engineering construct	Defensible but not from a primary source	Current Delta Drills 14-day half-life; the 0.30/0.60/0.85 alpha choices; the cold-start ladder 25/50/75
Anecdotal	Stated without primary-source backing	"Khan Academy moved away from streak-based mastery"

Mapping each model's components to this tier:

Model	Mathematical core	Educational application	Specific parameter values
Current EWMA	Textbook	Engineering	Engineering (chosen, not derived)

Model	Mathematical core	Educational application	Specific parameter values
Elo	Textbook	Peer-reviewed empirical (Pelánek 2016, 2017)	Engineering (K, decay schedule)
Streak	Trivial	Anecdotal / baseline	Engineering (threshold N)
Beta-Bernoulli	Textbook	Peer-reviewed novel proposal (Bijl 2025 only)	Engineering (decay half-life, gate thresholds)

7. Reading list for personal evaluation

In the audited corpus at [papers/mastery-estimation/](#) . Listed in priority order if you want to verify the most load-bearing claims:

1. [2016_pelanek_elo-adaptive-educational.pdf](#) — Pelánek's focused study of Elo in adaptive education. If you're considering Elo, this is the primary source. **Read this to verify the "r ≈ 0.70 after 5 responses" claim from Report 2** and to understand K-factor calibration choices.
2. [2017_pelanek_bkt-logistic-and-beyond.pdf](#) — Pelánek's broader survey. Less focused on Elo specifically but contextualizes it against BKT and logistic models. Hypothesis 1 (p. 33) is the load-bearing claim about when to prefer logistic vs BKT.
3. [2019_park-cornillie_multidim-irt-monitoring-ability.pdf](#) — Park 2019 / M-ERS. **Read this if multi-tag Elo aggregation is a concern.** Determines whether the upgrade is worth the complexity vs. just doing equal-credit averaging.
4. [2021_maier-baker-stalzer_pfa-challenges.pdf](#) — Maier 2021. **Already read in full.** Headline finding: even-skill averaging (what your current engine does for composites) AUC 0.7849 beat compensatory Performance Factors Analysis AUC 0.7818. Section 4.3 is the relevant comparison.
5. [2023_skycak_fire-spaced-repetition.html](#) — Math Academy's published architecture. **Already read in full.** Verbatim formulas for FIRE. Math Academy's deliberate rejection of Bayesian Knowledge Tracing is on the record here.
6. [2025_bijl_probabilistic-decay-trees.pdf](#) — Bijl 2025. **Recommended if Beta-Bernoulli is seriously considered.** The only published instance of a Beta-with-decay-over-graph mastery system. Single author, arXiv preprint, untested at scale — read critically.
7. [2017_pelanek_bkt-logistic-and-beyond.pdf](#) (same as 2 above) — also relevant for "why not BKT" rebuttals.

Not in the downloaded corpus but referenced and potentially worth retrieving: - **Yudelson 2019, "Elo, I Love You Won't You Tell Me Your K"** — the focused paper on K-factor calibration. - **Glickman, "Glicko-2"** — the standard extension adding rating deviation (forgetting) to Elo.

Documents in this series: - [Delta-Drills-Current-Model-EWMA.md](#) — complete specification of the engine deployed today - [Delta-Drills-Mastery-Models-Evidence-Review.md](#) — this

document - `papers/MASTERY_ESTIMATION_REFERENCE_v2.md` — audited reference (v2 after the 2026-05-24 source audit that flagged ~1/3 of v1 claims as wrong or fabricated)